

Appendix A: List of Campaigns Included

Beginning Year	Campaign	Location
2009	Frente Nacionalæde Resistencia Popular (FNRP)	Honduras
2009	Anti-Socialist Demonstrations	Bulgaria
2009	Moldovan "Grapevine" Revolution	Moldova
2009	2009 Anti-Saakashvili Campaign	Georgia
2009	anti-Rajoelina movement	Madagascar
2009	anti-Ravalomanana movement	Madagascar
2009	Green Revolution and Day of Rage	Iran
2009	Al Qaeda in the Arabian Peninsula	Yemen
2010	Snow Revolution	Russia
2010	Pro-Ouattara Campaign	Ivory Coast
2010	Renewed Western Sahara Independence Protests	Western Sahara
2010	Anti-Abdelaziz Bouteflika Campaign	Algeria
2010	Anti-Ben Ali Campaign (Jasmin Revolution)	Tunisia
2010	Anti-Interim Government	Kyrgyzstan
2010	Second Revolution	Kyrgyzstan
2010	Maoist Anti-Govt Protests	Nepal
2011	Anti-Aziz Protests	Mauritania
2011	Djibouti Arab Spring	Djibouti
2011	Libyan Civil War	Libya
2011	Day of Rage Protests	Iraq
2011	Syrian Uprising	Syria
2011	Syrian Civil War	Syria
2011	Protest for Constitutional Reform	Jordan
2011	Anti Ali Adduallah Saleh Campaign	Yemen
2011	Anti-King Hamad Campaign	Bahrain
2012	2012 Anti-Government Protests	Romania
2012	Northern Mali conflict	Mali
2012	Anti-Wade June 23 Movement	Senegal
2012	Let's Save Togo (Anti-Gnassingbe)	Togo
2012	M23	Democratic Republic of Congo
2012	Anti-Shiite Government Protests	Iraq
2012	Kirti Monastery Protests	China
2012	Anti-Nasheed Campaign	Maldives
2012	Nasheed Supporters	Maldives
2012	Anti-Maoist Campaign	Nepal
2013	Dance with me Campaign	Bulgaria
2013	Euromaidan	Ukraine
2013	anti-Balaka	Central African Republic
2013	Anti-Islamist Government Protests	Tunisia
2013	Protests Against the Crisis Talks	Tunisia
2013	Anti-Erdogan	Turkey
2013	Anti-Morsi Protests	Egypt

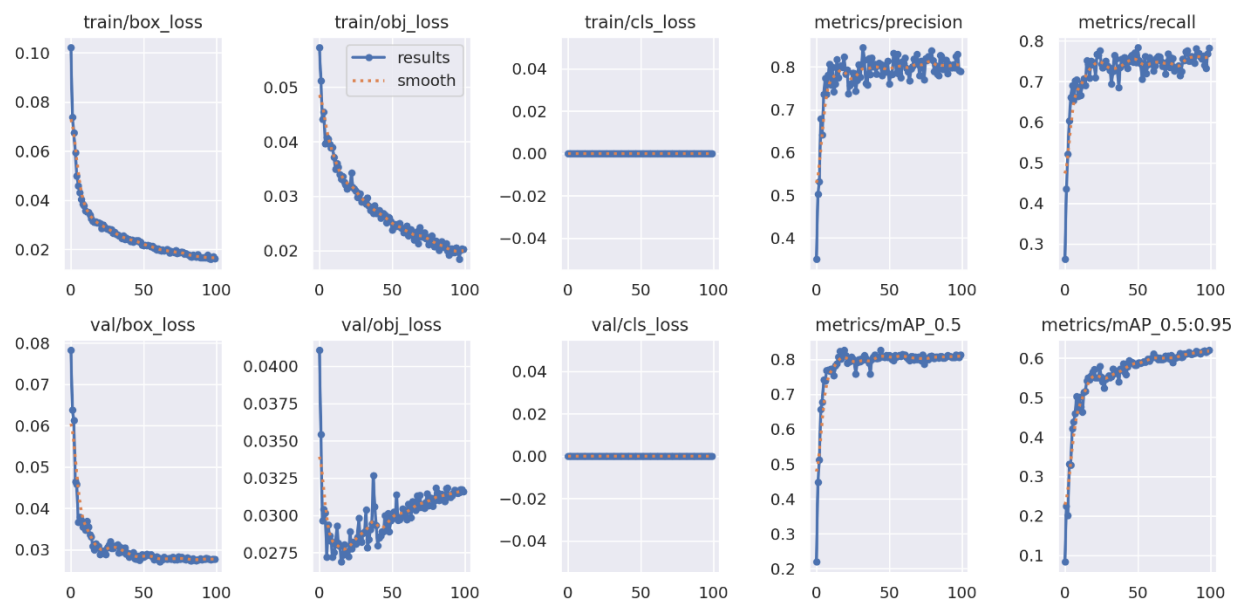
2013	Pro-Morsi Protests	Egypt
2013	Sinai Insurgency	Egypt
2013	Islamic State	Syria
2013	Civil Movement for Democracy	Thailand
2013	2013 Cambodian election protests	Cambodia
2014	Anti-Martelly campaign	Haiti
2014	2014 Mexican anti-Corruption Campaign	Mexico
2014	anti-Maduro	Venezuela
2014	UDAR	Bosnia-Herzegovina
2014	Ultras "United Ukraine" Campaign	Ukraine
2014	Rebels in East Ukraine (Ukrainian Civil War)	Ukraine
2014	Anti-Compaore Campaign	Burkina Faso
2014	Islamist militia factions (Second Libyan Civil War)	Libya
2014	Hong Kong pro-democracy (Umbrella Movement)	China
2014	New Democracy Movement	Thailand
2015	Guatemala Uprising	Guatemala
2015	Honduran indignados	Honduras
2015	Out Correa Campaign	Ecuador
2015	Anti-Rousseff	Brazil
2015	Montenegrin Opposition Protest	Montenegro
2015	Colorful Revolution	Macedonia
2015	Kosovo Anti-Government Protests	Kosovo
2015	Moldovan Anti-Government Protest	Moldova
2015	Colectiv Revolution	Romania
2015	Anti-Nkurunziza Campaign	Burundi
2015	Renewed Kurdistan Worker's Party (PKK)	Turkey
2015	Anti-Houthi Movement	Yemen
2015	ISIS-Afghanistan	Afghanistan
2015	Anti-Park protests	South Korea
2015	Western Southeast Asia Front	Myanmar
2015	Anti-Yameen	Maldives
2016	Catalan separatist protest	Spain
2016	Anti-right wing government	Poland
2016	Panama Papers Anti-Government Protest	Iceland
2017	Anti-Morales Protests	Guatemala
2017	Anti-Hernandez	Honduras
2017	Anti-Orban	Hungary
2017	Anti-Vucic	Serbia
2017	Romania anti-government protest	Romania
2017	2017 Anti-Gnassingbe	Togo
2017	anti-Kabila campaign	Democratic Republic of Congo
2017	2017 Iran Anti-Government Protests	Iran
2017	Anti-Erdogan II	Turkey
2018	Anti-Jovenel Moise Protests	Haiti

2018	Anti-Ortega Protests	Nicaragua
2018	Yellow Vests	France
2018	2018 Anti-government protests	Armenia
2019	Hernández Must Go campaign	Honduras
2019	anti-Morales protests	Bolivia
2019	anti-Áñez protests	Bolivia
2019	anti-Pinera	Chile
2019	Albanian anti-corruption	Albania
2019	97000 Resist campaign	Montenegro
2019	2019 Georgian Protests	Georgia
2019	Smile Revolution	Algeria
2019	2019 Iranian protests	Iran
2019	anti-Mahdi movement	Iraq
2019	anti-Sisi protests	Egypt
2019	October Revolution	Lebanon

Appendix B: Object Detection Algorithm Training Discussion

Figure B1 presents the plots of interest for the fine-tuning process of the YOLO v5 for the protest banner detection task. As mentioned in the paper, object detection task requires the model to first, locate the targeted objects, and second, classify the detected objects. Correspondingly, two loss functions are calculated during the model training: box loss (for the location task) and object loss (for the classification task). As shown in the plots, while the box validation loss keeps decreasing, the object validation loss starting increase during the training, which suggests the potential problem of overfitting. Accordingly, for the final algorithm to use, we choose the one with highest [mAP@0.5](#) and [mAP@0.95](#) values instead of the fine-tuned model output at the 100 epochs. And the final evaluation statistics is shown in Table B1 and the evaluation plots are shown in Figure B2.

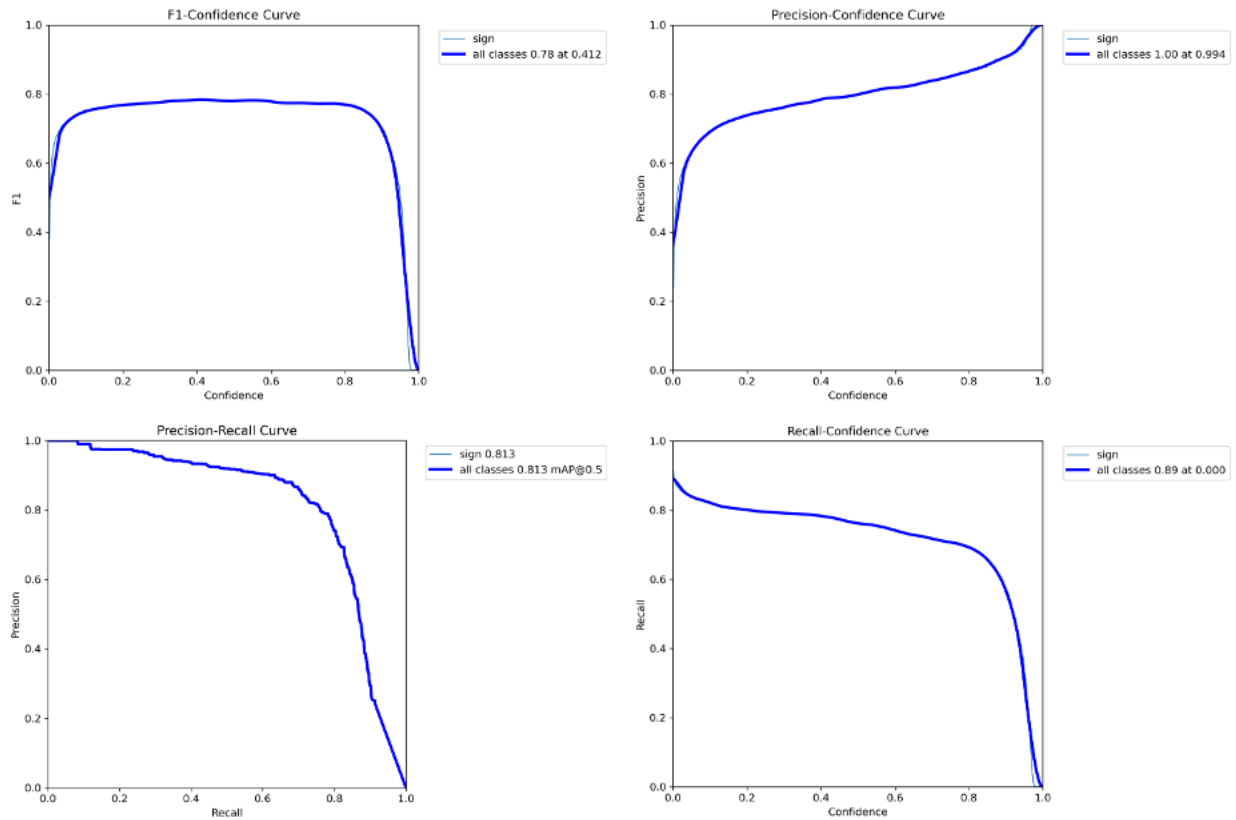
Figure B1: Training results



Class	Instances	Precision (P)	Recall (R)	mAP@0.5	mAP@0.5:0.95
Protest Sign	781	0.789	0.781	0.813	0.620

Table B1: Best model performance

Figure B2: Evaluation Plots



One of the reasonable concerns shared across the computational social science community is whether the performance of the after-training algorithm is good enough for downstream substantive tasks. Different from many tasks in the computer science field in which most experiments are finished in a laboratory environment with well-designed datasets (or benchmarks), social scientists who want to utilize computational tools often face much messier data. Accordingly, a sophisticated balance is needed for both allowing the social scientists to take advantage of the advanced AI tools as well as ensuring the credibility of the empirical findings from the AI-coded data. In this Appendix, we try to convince the readers that, given the complicated nature of the protest photos online, our choices of computational tools yield enough credibility for the final empirical findings.

Figure B3 provides a straightforward way to visualize the complicated nature lying in the coded dataset. We manually coded 1000 protest photos randomly sampled from the collected database. The left label correlogram plots the information of the ground truth bounding boxes in the coded dataset. The lower right subplot shows that most protest banners annotated are rather small. On the right plot, the subplots on the upper right (all the bounding boxes) and lower left (frequencies of the bounding boxes at each position) suggest that the protest banners could appear in any corners in one photo. The small scale of the banners in photos and the random positions all make the detection task even more challenging. To make it worse, there are many “noisy” photos with low resolution, as shown in Figure A4, in the big dataset, which makes the protest banner detection task in those photos almost impossible.

Figure B3: Distributions of the ground truth bounding boxes

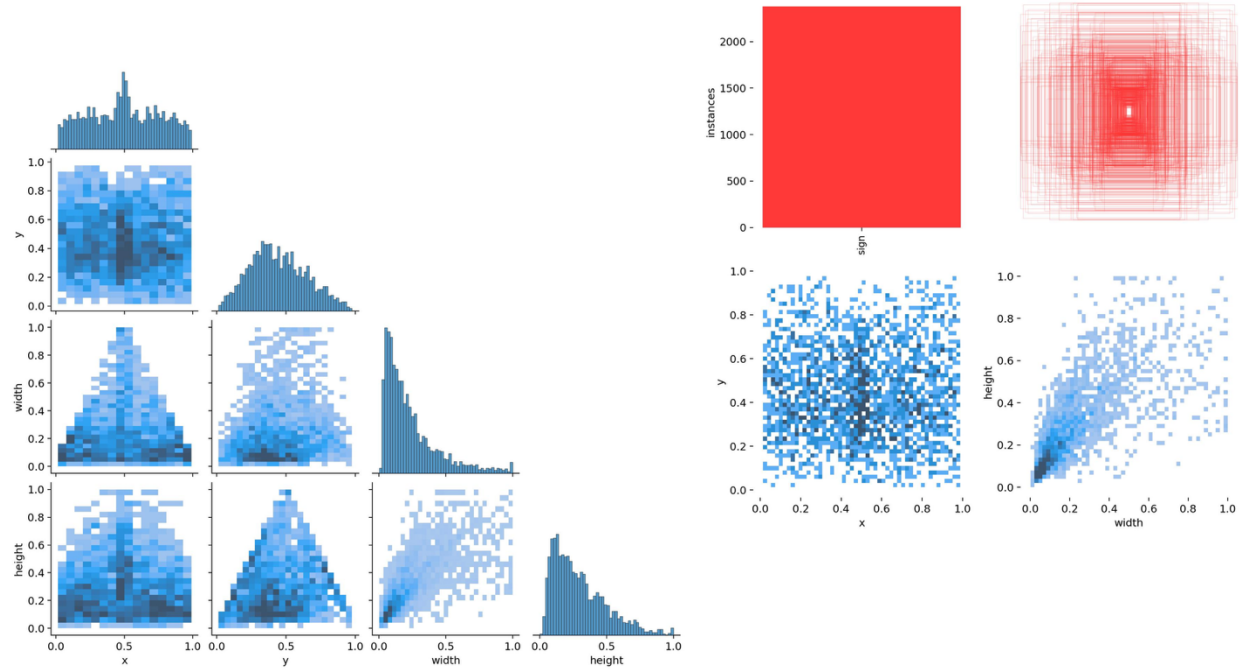


Figure B4: Noisy photos



Despite these challenges, careful training design enabled our algorithm to reach around 0.82 $mAP@0.5$ —state of the art in 2022, when the study was conducted—and performance comparable to other CV-based detectors of the period (Rekavandi et al. 2023) and even to systems used in advanced real-world applications such as autonomous driving (Balasubramaniam & Pasricha 2022). Furthermore, through manual checking on the randomly sampled predictions, the model shows pretty good face validity.

For more rigorous test on the validation of the object detection algorithm, we constructed a small set ($n=50$) of protest photos with complicated features but high resolution. Figure B5 shows some of the sample predictions. And the $mAP@50$ statistic arrives 0.863 with the precision score at 0.902, which is comparable to the performance of more advanced YOLO models that training on much larger datasets.

Figure B5: Face validation of the object detection algorithm on complicated high-resolution protest photos

Appendix C: Statistical Analysis

C.I. Curvilinear Hypothesis Appendix (Hypothesis 1)

C.A1. Strict-success and ordered-outcome models

Appendix Table C1: Additional Robustness Models for the Curvilinear Hypothesis: Effect of English-Language Signs on Strict Campaign Success and Ordered Campaign Outcomes, 2009-2019

Weibull accelerated-failure-time (AFT) models for time-to-event outcomes; Firth penalized maximum likelihood for binary success outcomes; adjacent-category ordered logit (Kosmidis-Firth penalty) for ordered campaign outcomes. All annual-campaign Cox proportional-hazards models use counting-process format with standard errors clustered by source campaign.

Columns 1-3 use NAVCO 1.3 campaign-level data: time to strict success, strict-success binary, and ordered campaign success with three categories. Columns 4-6 use NAVCO 2.1 campaign-year panel data: hazard of strict success, yearly strict-success binary, and ordered annual progress with six levels. These specifications complement the main concession-only models by varying the success threshold and outcome scale. Events per variable (EPV) is reported in the table footer; an asterisk marks EPV below 5. EPV is not defined for ordered adjacent-category logit models.

Response	Time to Strict Success	Strict Success (Binary)	Ordered Campaign Success	Hazard of Strict Success	Strict Success Yearly (Binary)	Ordered Annual Progress
Unit of Analysis	Campaign	Campaign	Campaign	Campaign-year	Campaign-year	Campaign-year
Model choice	Weibull AFT	Firth PML	bracl	Cox PH	Firth PML	bracl
Proportion of English-Language Signs Squared	186.395* (100.433)	-238.239 (212.792)	128.843 (116.083)	-338.445 (327.122)	-143.345 (263.436)	27.947 (59.428)
Proportion of English-Language Signs	-37.856* (21.312)	49.657 (47.155)	-25.933 (25.739)	78.131 (68.221)	38.726 (58.855)	-8.177 (13.615)
Campaign Primarily Nonviolent	-0.176 (0.478)	-0.855 (0.863)	0.303 (0.470)	0.599 (1.176)	0.419 (0.923)	0.039 (0.223)
Campaign Purpose Regime Change	-10.387 (3286.533)	2.170* (1.422)	-0.950 (0.673)	19.282*** (0.707)	1.526 (1.244)	-0.226 (0.250)
Campaign Location V-Dem CSO Repression	-0.376*** (0.106)	0.382* (0.206)	-0.258** (0.115)	0.922** (0.403)	0.766** (0.381)	-0.219** (0.093)
Campaign Length, in Years (ln)		-0.989** (0.456)	0.381 (0.239)			
Time in Campaign					-0.058 (0.427)	0.108 (0.101)
Number of Images on Campaign (ln)	0.304*** (0.107)	-0.089 (0.232)	0.042 (0.127)			
Number of Images in Campaign-Year (ln)				0.424 (0.419)	0.347 (0.404)	-0.061 (0.105)
Campaign Location Population (ln)	-0.037 (0.077)	0.196 (0.164)	-0.089 (0.090)			
Campaign-Year Population (ln)				0.327 (0.303)	0.224 (0.267)	-0.072 (0.071)
English Colonial Legacy	0.042 (0.331)	0.345 (0.627)	-0.307 (0.343)	1.003 (1.025)	0.871 (0.911)	-0.335 (0.227)
Constant	12.885	-6.704			-11.245*	

	(3286.533)	(4.095)		(5.796)	
Observations	100	100	100	65	65
EPV (* = <5)	3.7*	3.3*		1.0*	0.8*

*** p<0.01, ** p<0.05, * p<0.1

C.A2. Distributional and clustering sensitivity

Appendix Table C2: Distributional and Clustering Sensitivity for the Curvilinear Hypothesis: Effect of English-Language Signs on Concession Outcomes Under Alternative Model Specifications, 2009-2019

Sensitivity checks for the inverted-U curvature claim. Column 1 replaces the Weibull baseline with log-normal accelerated-failure-time, addressing concern about distributional leverage of the Weibull functional form. Column 2 drops the parametric baseline assumption entirely with a semi-parametric Cox proportional-hazards model on the NAVCO 1.3 campaign-level sample. Column 3 re-fits the NAVCO 2.1 first-event Cox specification using Andersen-Gill recurrent-event format, retaining repeated within-campaign events instead of truncating after the first concession. Column 4 re-fits the NAVCO 2.1 yearly Firth penalized maximum likelihood model using a generalized estimating equation logit with exchangeable working correlation and source-campaign clustering.

These specifications are robustness checks of individual main-panel models, not additional hypothesis tests, and therefore lie outside the family-wise multiplicity correction applied in the main table. Events per variable (EPV) is reported in the footer; an asterisk marks EPV below 5.

Response	Time to Concession (log-normal)	Hazard of Concession (Cox PH)	Recurrent Concession Hazard	Yearly Concession (clustered)
Unit of Analysis	Campaign	Campaign	Campaign-year	Campaign-year
Model choice	log-normal AFT	Cox PH	Cox PH (Andersen-Gill)	GEE logit
Proportion of English-Language Signs Squared	123.788 (82.445)	-212.087 (147.097)	-282.965 (176.689)	-552.027** (271.819)
Proportion of English-Language Signs	-23.294 (18.289)	42.651 (31.763)	75.647* (41.102)	131.507** (63.470)
Campaign Primarily Nonviolent	-0.101 (0.307)	0.306 (0.710)	-0.019 (0.591)	0.043 (0.760)
Campaign Purpose Regime Change	-1.014** (0.471)	1.584 (1.066)	1.553** (0.759)	1.785** (0.752)
Campaign Location V-Dem CSO Repression	-0.302*** (0.079)	0.580*** (0.168)	1.027*** (0.327)	1.207*** (0.401)
Time in Campaign				-0.167 (0.376)
Number of Images on Campaign (ln)	0.245*** (0.080)	-0.390** (0.163)		
Number of Images in Campaign-Year (ln)			-0.048 (0.345)	-0.071 (0.544)
Campaign Location Population (ln)	-0.007 (0.061)	0.089 (0.114)		
Campaign-Year Population (ln)			0.250 (0.232)	0.149 (0.377)
English Colonial Legacy	-0.019	0.419	2.051***	2.464***

	(0.229)	(0.448)	(0.774)	(0.883)
Constant	2.003			-12.895**
	(1.471)			(5.539)
Observations	100	100	65	65
EPV (* = <5)	5.0	5.6	1.9*	1.5*

*** p<0.01, ** p<0.05, * p<0.1

Part II. Political Vulnerability Models (Hypothesis 2.1)

C.A1. Additional H2.1 ordered-outcome and robustness models

Appendix Table C3: Additional H2.1 Ordered-Outcome and Robustness Models — The Role of English-Language Signs, Conditional on Political Vulnerability to External Pressure, on Campaign Progress and Concession, 2009-2019

Adjacent-category ordered logit models are used for ordered campaign success and ordered annual progress. Recurrent-event Cox models are used for repeated concession hazards, and clustered GEE logit models are used for yearly concession outcomes. Standard errors for annual campaign models are clustered by NAVCO 2.1 source campaign; all interaction terms use mean-centered constituent variables.

Columns 1-4 report ordered-outcome extensions for campaign-level success and annual campaign progress. Columns 5-8 report annual campaign robustness checks using recurrent-event Cox models and clustered GEE logit models. These models are additional appendix diagnostics rather than the confirmatory main-table specifications. EPV means events per variable; an asterisk marks EPV below 5. EPV is not defined for ordered adjacent-category logit models.

Response	Ordered Success	Ordered Success	Ordered Progress	Ordered Progress	Recurrent Concession Hazard	Recurrent Concession Hazard	Yearly Concession	Yearly Concession
Unit of Analysis Model choice	Campaign bracl	Campaign bracl	Campaign-year bracl	Campaign-year bracl	Campaign-year Cox PH	Campaign-year Cox PH	Campaign-year GEE logit	Campaign-year GEE logit
Proportion of English-Language Signs	1.687 (4.344)	1.419 (4.433)	-2.378 (2.660)	-1.848 (2.522)	15.886 (10.965)	11.856 (8.515)	11.631 (10.001)	9.819 (9.179)
Domestic Autonomy	0.105 (0.158)		0.248* (0.144)		-1.099*** (0.408)		-1.271*** (0.421)	
Proportion English-Language Signs X Domestic Autonomy	-7.182 (5.305)		-2.830 (2.880)		17.946* (10.890)		16.295 (11.998)	
State Authority Over Territory		0.010 (0.012)		0.012 (0.010)		-0.046* (0.024)		-0.057 (0.037)
Proportion English-Language Signs X State Authority Over Territory		-0.285 (0.319)		-0.065 (0.163)		0.341 (0.621)		0.354 (0.550)
Campaign Length, in Years (ln)	0.393* (0.231)	0.405* (0.235)						
Time in Campaign (years)			0.084 (0.104)	0.087 (0.099)			-0.077 (0.292)	-0.075 (0.252)
Campaign Primarily Nonviolent	0.276	0.092	-0.129	-0.185	0.809	0.946	1.007	1.244

	(0.512)	(0.566)	(0.264)	(0.313)	(0.505)	(0.625)	(0.683)	(0.881)
Campaign Purpose Regime Change	-0.816	-0.767	-0.097	-0.059	0.766	0.462	0.596	0.556
	(0.667)	(0.660)	(0.238)	(0.235)	(0.976)	(0.986)	(0.914)	(0.799)
Campaign Location V-Dem CSO Repression	-0.208*	-0.210*	-0.239***	-0.177**	1.137***	0.766***	1.227***	0.832**
	(0.108)	(0.107)	(0.093)	(0.082)	(0.392)	(0.283)	(0.381)	(0.377)
Number of Images on Campaign (ln)	-0.006	-0.025						
	(0.121)	(0.121)						
Number of Images in Campaign-Year (ln)			-0.155	-0.124	0.508*	0.375	0.547	0.388
			(0.103)	(0.098)	(0.308)	(0.305)	(0.377)	(0.412)
Constant							-5.313**	-4.523*
							(2.249)	(2.315)
Observations	101	101	65	65	65	65	65	65
EPV (* = <5)					2.1*	2.1*	1.7*	1.7*

*** p<0.01, ** p<0.05, * p<0.1

Part III. Economic Vulnerability Models (Hypothesis 2.2)

C.A1. Additional H2.2 ordered-outcome and robustness models

H22 Appendix Rest: Ordered Outcome and Campaign-Year Robustness Models

Appendix Table C4: Ordered Outcome Models

Response	Ordered Success	Ordered Progress	Ordered Success	Ordered Progress	Ordered Success	Ordered Progress	Ordered Success	Ordered Progress
Unit of Analysis	Campaign	Campaign- Year	Campaign	Campaign- Year	Campaign	Campaign- Year	Campaign	Campaign- Year
Model choice	Adjacent-cat. logit	Adjacent-cat. logit	Adjacent-cat. logit	Adjacent- cat. logit	Adjacent- cat. logit	Adjacent- cat. logit	Adjacent- cat. logit	Adjacent- cat. logit
Proportion of English-Language Signs	3.454	-1.912	1.442	-3.877	1.659	0.521	3.823	-1.207
	(4.612)	(2.694)	(4.934)	(3.012)	(6.129)	(3.044)	(7.254)	(2.728)
Campaign Location FDI Inflows (% of GDP)	0.077*	0.016						
	(0.040)	(0.029)						
Proportion English-Language Signs X FDI Inflows (% of GDP)	-0.479	0.816						
	(1.231)	(0.667)						
International Trade Freedom			0.280**	-0.232*				
			(0.133)	(0.128)				

Proportion English-Language Signs X International Trade Freedom			-6.813*	0.226				
			(3.810)	(2.159)				
Foreign Aid (ln)					-0.064	-0.076		
					(0.163)	(0.078)		
Proportion English-Language Signs X Foreign Aid (ln)					-9.129	-3.460*		
					(6.243)	(2.090)		
Number of IO Donors							-0.032	0.017
							(0.079)	(0.033)
Proportion English-Language Signs X Number of IO Donors							-7.084**	-0.041
							(3.084)	(0.886)
Campaign Length / Time in Campaign	0.336	0.112	0.354	0.063	0.815**	0.103	1.205**	0.085
	(0.245)	(0.103)	(0.265)	(0.112)	(0.401)	(0.115)	(0.501)	(0.118)
Campaign Primarily Nonviolent	-0.058	-0.001	-0.273	-0.169	0.067	-0.017	0.523	-0.088
	(0.504)	(0.258)	(0.620)	(0.307)	(0.669)	(0.264)	(0.742)	(0.266)
Campaign Purpose Regime Change	-0.858	-0.149	-0.753	-0.300	-0.674	-0.038	-1.276	0.057
	(0.665)	(0.243)	(0.718)	(0.290)	(0.879)	(0.273)	(1.129)	(0.247)
Campaign Location V-Dem CSO Repression	-0.250**	-0.149*	-0.445***	-0.093	-0.313	-0.245**	-0.251	-0.206**
	(0.117)	(0.090)	(0.150)	(0.102)	(0.209)	(0.099)	(0.215)	(0.092)
Number of Images on Campaign / in Campaign-Year (ln)	0.005	-0.131	-0.185	-0.181	-0.207	-0.233*	-0.204	-0.221*
	(0.141)	(0.108)	(0.151)	(0.114)	(0.246)	(0.134)	(0.237)	(0.121)
Campaign Location GDP / Campaign-Year GDP per Capita (ln)	0.163	0.005	0.136	0.153	0.075	0.090	0.100	0.155
	(0.136)	(0.108)	(0.161)	(0.147)	(0.232)	(0.126)	(0.304)	(0.150)
Constant								
Observations	97	61	91	60	44	59	44	59

Adjacent-category ordered logit models are used for ordered campaign success and ordered annual campaign progress. Standard errors are reported in parentheses, and all interaction terms use mean-centered constituent variables. Campaign-level ordered success is NAVCO 1.3 success3; campaign-year ordered progress is NAVCO 2.1 progress_ord with failure separated.

Appendix Table C5: Campaign-Year Robustness Models

						(4.105)	(5.566)	
Time in Campaign (years)		-0.152		0.120		-0.088		0.191
		(0.261)		(0.296)		(0.274)		(0.441)
Campaign Primarily Nonviolent	0.083	0.015	1.396*	1.068	0.692	0.800	0.736	0.953
	(0.711)	(0.821)	(0.784)	(0.823)	(0.935)	(1.140)	(0.966)	(1.072)
Campaign Purpose Regime Change	0.704	0.780	1.950*	2.052*	-0.277	-0.130	-0.398	0.051
	(0.818)	(0.799)	(1.101)	(1.201)	(0.875)	(1.036)	(0.751)	(0.921)
Campaign Location V-Dem CSO Repression	0.608**	0.734**	0.614	0.487	0.892***	1.169***	0.965**	1.228**
	(0.241)	(0.341)	(0.374)	(0.478)	(0.282)	(0.408)	(0.385)	(0.533)
Number of Images in Campaign-Year (ln)	0.387	0.385	1.017***	0.748*	0.786*	1.027	0.945*	1.223*
	(0.325)	(0.390)	(0.384)	(0.453)	(0.435)	(0.629)	(0.532)	(0.732)
Campaign-Year GDP per Capita (ln)	-0.049	0.052	-0.901**	-0.782	-0.491	-0.640	-0.551	-0.935
	(0.379)	(0.453)	(0.437)	(0.519)	(0.515)	(0.644)	(0.524)	(0.684)
Constant		-3.917		-1.966		-2.160		-1.487
		(3.198)		(4.196)		(3.461)		(4.885)
Observations	61	61	60	60	59	59	59	59

Recurrent-event Cox proportional-hazards models and campaign-clustered GEE logit models are used for annual broad-success/concession robustness checks. Standard errors are reported in parentheses, and all interaction terms use mean-centered constituent variables.

These robustness rows are outside the confirmatory Table 3 multiplicity family and should be read as appendix diagnostics.

*** p<0.01, ** p<0.05, * p<0.1

Part IV. Domestic / Internal Vulnerability Models (Hypothesis 3)

C.A1. Ordered-outcome models on campaign success and annual progress

Appendix Table C6: Ordered-Outcome Models for the Domestic Vulnerability Hypothesis: Adjacent-Category Logit on Campaign Success and Annual Progress, Conditional on Domestic Institutional Openness, 2009-2019

Adjacent-category ordered logit with parallel slopes (Kosmidis-Firth penalty) on two outcomes: campaign-level three-category success on NAVCO 1.3 (columns 1-4) and annual progress on a six-level scale with failure coded separately from status quo on NAVCO 2.1 (columns 5-8). Both panels test the same four moderators of domestic institutional openness. When CSO Repression serves as the moderator, the domestic control swaps to Polity Score on NAVCO 1.3 and to none on NAVCO 2.1, avoiding moderator-control collinearity. Category-specific intercepts are omitted from the table; reported coefficients are the shared parallel-slope predictor effects.

All rows are adjusted with the Benjamini-Hochberg correction within the appendix-grade family. Events per variable is not defined for ordered adjacent-category logit models.

Response	Success (3-cat, NAVCO 1.3)	Success (3-cat, NAVCO 1.3)	Success (3-cat, NAVCO 1.3)	Success (3-cat, NAVCO 1.3)	Progress (6-level, NAVCO 2.1)	Progress (6-level, NAVCO 2.1)	Progress (6-level, NAVCO 2.1)	Progress (6-level, NAVCO 2.1)
Moderator	CSO Repression	Civil Liberties	Political Liberty	Polity	CSO Repression	Civil Liberties	Political Liberty	Polity
Model choice	bracl	bracl	bracl	bracl	bracl	bracl	bracl	bracl
Proportion of English-Language Signs (centered)	4.563 (5.021)	3.447 (4.633)	4.479 (4.908)	6.047 (5.094)	-0.744 (2.687)	-0.804 (2.788)	-0.438 (2.778)	-1.678 (2.825)
Campaign Location V-Dem CSO Repression (centered)	-0.392** (0.159)				-0.180** (0.087)			
Campaign Location V-Dem Civil Liberties (centered)		0.152 (1.037)				1.201 (1.121)		
Campaign Location V-Dem Political Liberty (centered)			-0.652 (1.193)				1.065 (1.326)	
Campaign Location Polity Score (centered)				0.044 (0.037)				0.031 (0.027)
Eng x CSO Repression	-5.612 (4.114)				-2.152 (2.326)			
Eng x Civil Liberties		-31.831 (22.149)				-17.074 (14.485)		
Eng x Political Liberty			-34.246 (23.583)				-13.544 (13.605)	
Eng x Polity Score				-2.324** (1.171)				-0.575 (0.578)
Campaign Primarily Nonviolent	-0.150 (0.516)	0.033 (0.523)	0.013 (0.507)	0.061 (0.534)	-0.012 (0.265)	-0.102 (0.275)	-0.019 (0.261)	-0.019 (0.279)
Campaign Purpose Regime Change	-0.690 (0.675)	-0.775 (0.682)	-0.781 (0.688)	-0.783 (0.710)	-0.073 (0.235)	-0.026 (0.244)	-0.046 (0.240)	-0.120 (0.265)
Polity Score (control)	0.046 (0.038)							
CSO Repression (control)		-0.264 (0.191)	-0.127 (0.235)	-0.405** (0.161)		-0.361* (0.193)	-0.364 (0.261)	-0.301** (0.130)
Time in Campaign (years)					0.099 (0.103)	0.074 (0.103)	0.086 (0.102)	0.107 (0.105)
Campaign Length, in Years (ln)	0.538** (0.256)	0.409* (0.243)	0.425* (0.246)	0.666** (0.275)				
Number of Images on Campaign (ln)	-0.148 (0.145)	-0.102 (0.135)	-0.095 (0.134)	-0.261 (0.163)	-0.135 (0.102)	-0.151 (0.106)	-0.154 (0.106)	-0.226* (0.120)

Campaign Location GDP per capita (ln, 1.3)	0.150	0.116	0.120	0.171				
	(0.142)	(0.133)	(0.132)	(0.145)				
Campaign-Year GDP per capita (ln, 2.1)					0.025	0.055	0.089	-0.015
					(0.105)	(0.107)	(0.126)	(0.120)
Observations	94	98	98	94	64	64	64	62
EPV (* = <5)								

*** p<0.01, ** p<0.05, * p<0.1

C.A2. Recurrent-event and clustering sensitivity

Appendix Table C7: Recurrent-Event and Clustering Sensitivity for the Domestic Vulnerability Hypothesis: Andersen-Gill Recurrent-Event Cox and Source-Clustered GEE Logit Companion Models on NAVCO 2.1 Panel, 2009-2013

Sensitivity checks for the campaign-year results. Columns 1-4 re-fit the Cox proportional-hazards concession models using Andersen-Gill recurrent-event format on the full panel (no first-event truncation). Columns 5-8 re-fit the yearly Firth concession models using generalized estimating equation logit with exchangeable working correlation and source-campaign clustering.

These specifications are robustness checks of individual main-panel models, not additional hypothesis tests, and therefore lie outside the family-wise multiplicity correction. Events per variable (EPV) is reported in the footer; an asterisk marks EPV below 5.

Response	Concession Hazard (Recurrent)	Concession Hazard (Recurrent)	Concession Hazard (Recurrent)	Concession Hazard (Recurrent)	Yearly Concession (GEE clustered)	Yearly Concession (GEE clustered)	Yearly Concession (GEE clustered)	Yearly Concession (GEE clustered)
Moderator	CSO Repression	Civil Liberties	Political Liberty	Polity	CSO Repression	Civil Liberties	Political Liberty	Polity
Model choice	Cox PH (AG)	Cox PH (AG)	Cox PH (AG)	Cox PH (AG)	GEE logit	GEE logit	GEE logit	GEE logit
Proportion of English-Language Signs (centered)	3.203	6.319	3.953	7.252	-1.214	4.695	-0.716	-3.254
	(8.926)	(8.559)	(9.213)	(10.468)	(10.638)	(10.518)	(10.443)	(13.844)
Campaign Location V-Dem CSO Repression (centered)	0.715***				0.944**			
	(0.265)				(0.446)			
Campaign Location V-Dem Civil Liberties (centered)		-5.394				-5.286		
		(4.175)				(4.988)		
Campaign Location V-Dem Political Liberty (centered)			-5.474				-6.334	
			(3.691)				(4.930)	
Campaign Location Polity Score (centered)				-0.151**				-0.362***
				(0.068)				(0.120)
Eng x CSO Repression	9.968*				13.932			
	(6.052)				(9.947)			
Eng x Civil Liberties		65.764				65.060		
		(45.834)				(69.079)		

Eng x Political Liberty			52.135*				75.454	
			(30.789)				(62.454)	
Eng x Polity Score				3.230				6.468*
				(2.019)				(3.717)
Campaign Primarily Nonviolent	0.267	0.598	0.345	0.615	0.202	0.607	0.223	-0.043
	(0.844)	(0.696)	(0.681)	(0.538)	(0.913)	(0.856)	(0.840)	(1.049)
Campaign Purpose Regime Change	0.379	0.351	0.299	0.484	0.398	0.443	0.274	0.335
	(0.789)	(0.798)	(0.816)	(0.992)	(0.846)	(0.792)	(0.888)	(1.280)
CSO Repression (control)		1.513**	1.657**	1.206***		1.662*	2.026**	2.601***
		(0.717)	(0.701)	(0.261)		(0.895)	(1.020)	(0.711)
Time in Campaign (years)					-0.112	0.016	-0.076	-0.300
					(0.254)	(0.246)	(0.237)	(0.375)
Number of Images in Campaign-Year (ln)	0.329	0.389	0.452	0.782**	0.379	0.440	0.518	1.428***
	(0.300)	(0.290)	(0.350)	(0.320)	(0.399)	(0.418)	(0.487)	(0.520)
Campaign-Year GDP per capita (ln)	-0.030	-0.171	-0.300	0.087	0.066	-0.126	-0.285	0.627
	(0.384)	(0.398)	(0.432)	(0.348)	(0.440)	(0.485)	(0.593)	(0.649)
Constant					-4.101	-3.298	-1.744	-12.949**
					(3.120)	(3.320)	(3.845)	(5.686)
Observations	64	64	64	62	64	64	64	62
EPV (* = <5)	2.1*	1.9*	1.9*	1.9*	1.7*	1.5*	1.5*	1.5*

*** p<0.01, ** p<0.05, * p<0.1

C.A3. Additional V-Dem domestic-openness moderators

Appendix Table C8: Additional Domestic-Openness Moderators for the Domestic Vulnerability Hypothesis: Political Civil Liberties and Freedom of Association Companion Models, 2009-2019

Standalone extension to the main domestic-vulnerability table, testing two additional V-Dem moderators of domestic institutional openness: Political Civil Liberties (columns 1-2, included in the main panel as a parity check) and Freedom of Association thick index (columns 3-4, new). Columns 1 and 3 use Weibull accelerated-failure-time on time to broad concession; columns 2 and 4 use Firth penalized maximum likelihood on the binary broad-concession outcome. All interaction terms use mean-centered constituent variables.

These models extend the moderator set beyond the four moderators in the main panel (CSO Repression, Civil Liberties, Political Liberty, Polity Score). Controls are suppressed from the table for compactness. Events per variable (EPV) is reported in the footer; an asterisk marks EPV below 5.

Response	Time to Concession	Concession (Binary)	Time to Concession	Concession (Binary)
Moderator	Political Liberty	Political Liberty	Freedom of Association	Freedom of Association
Model choice	Weibull AFT	Firth PML	Weibull AFT	Firth PML
Proportion of English-Language Signs (centered)	4.848	-10.492	4.282	-10.920
	(3.533)	(8.252)	(3.459)	(8.356)

V-Dem Political Liberty (centered)	-0.997 (0.954)	0.723 (2.025)		
V-Dem Freedom of Association thick (centered)			-1.255 (0.898)	0.078 (1.852)
Eng x Political Liberty	-28.758* (16.229)	83.809** (41.259)		
Eng x Freedom of Association			-25.350 (17.118)	86.830** (43.144)
Observations	98	98	98	98
EPV (* = <5)	5.0	4.5*	5.0	4.5*

*** p<0.01, ** p<0.05, * p<0.1

Part V. General Robustness Tables

Appendix Tables C9–C10: Robustness Checks

These tables summarize additional robustness and sensitivity checks for the main specifications.

Appendix Table C9: Strict-success response robustness

Stars and p-values use the adjusted p-values already exported by appendix_tables.r.

Family	Model	Estimator	Dataset	Response variable	Moderator	Beta	Adjusted p	Correction	N
H1	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: English share squared	Weibull AFT	NAVCO 1.3	Time to success (strict)	English share squared	186.395	0.238	BH	100
H1	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: English share squared	Firth PML	NAVCO 1.3	Success (strict, binary)	English share squared	-238.239	0.415	BH	100
H1	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: English share squared	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	English share squared	-338.445	0.415	BH	65
H1	Firth PML; NAVCO 2.1; Success (CY, strict, bin.); Moderator: English share squared	Firth PML	NAVCO 2.1	Success (CY, strict, bin.)	English share squared	-143.345	0.653	BH	65
H2.1	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: Domestic autonomy	Weibull AFT	NAVCO 1.3	Time to success (strict)	Domestic autonomy	-6.068	0.892	BH	101
H2.1	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: State territorial control	Weibull AFT	NAVCO 1.3	Time to success (strict)	State territorial control	-0.120	1.000	BH	101
H2.1	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: Domestic autonomy	Firth PML	NAVCO 1.3	Success (strict, binary)	Domestic autonomy	11.044	0.892	BH	101
H2.1	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: State territorial control	Firth PML	NAVCO 1.3	Success (strict, binary)	State territorial control	0.276	1.000	BH	101
H2.1	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: Domestic autonomy	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	Domestic autonomy	1.000	1.000	BH	65

H2.1	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: State territorial control	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	State territorial control	0.032	1.000	BH	65
H2.1	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: Domestic autonomy	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	Domestic autonomy	1.629	1.000	BH	65
H2.1	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: State territorial control	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	State territorial control	-0.000	1.000	BH	65
H2.2	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: FDI inflows	Weibull AFT	NAVCO 1.3	Time to success (strict)	FDI inflows	-2.787	0.230	BH	97
H2.2	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: Trade openness	Weibull AFT	NAVCO 1.3	Time to success (strict)	Trade openness	-5.464	0.366	BH	91
H2.2	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: Foreign aid	Weibull AFT	NAVCO 1.3	Time to success (strict)	Foreign aid	1.871	0.904	BH	44
H2.2	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: IO donor count	Weibull AFT	NAVCO 1.3	Time to success (strict)	IO donor count	0.129	0.960	BH	44
H2.2	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: FDI inflows	Firth PML	NAVCO 1.3	Success (strict, binary)	FDI inflows	1.412	0.750	BH	97
H2.2	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: Trade openness	Firth PML	NAVCO 1.3	Success (strict, binary)	Trade openness	6.877	0.393	BH	91
H2.2	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: Foreign aid	Firth PML	NAVCO 1.3	Success (strict, binary)	Foreign aid	17.607	0.373	BH	44
H2.2	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: IO donor count	Firth PML	NAVCO 1.3	Success (strict, binary)	IO donor count	13.188	0.366	BH	44
H2.2	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: FDI inflows	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	FDI inflows	-10.464	0.373	BH	61
H2.2	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: Trade openness	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	Trade openness	18.762	0.373	BH	60
H2.2	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: Foreign aid	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	Foreign aid	19.614	0.373	BH	59
H2.2	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: IO donor count	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	IO donor count	-12.039	0.230	BH	59
H2.2	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: FDI inflows	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	FDI inflows	-5.203	0.373	BH	61
H2.2	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: Trade openness	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	Trade openness	-11.135	0.855	BH	60
H2.2	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: Foreign aid	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	Foreign aid	14.742	0.373	BH	59
H2.2	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: IO donor count	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	IO donor count	-9.558	0.373	BH	59
H3	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: CSO repression	Weibull AFT	NAVCO 1.3	Time to success (strict)	CSO repression	-2.859	0.703	BH	94
H3	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: Civil liberties	Weibull AFT	NAVCO 1.3	Time to success (strict)	Civil liberties	-12.650	0.737	BH	98
H3	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: Political liberty	Weibull AFT	NAVCO 1.3	Time to success (strict)	Political liberty	-15.793	0.717	BH	98

H3	Weibull AFT; NAVCO 1.3; Time to success (strict); Moderator: Polity	Weibull AFT	NAVCO 1.3	Time to success (strict)	Polity	0.356	0.778	BH	94
H3	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: CSO repression	Firth PML	NAVCO 1.3	Success (strict, binary)	CSO repression	2.948	0.777	BH	94
H3	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: Civil liberties	Firth PML	NAVCO 1.3	Success (strict, binary)	Civil liberties	25.668	0.737	BH	98
H3	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: Political liberty	Firth PML	NAVCO 1.3	Success (strict, binary)	Political liberty	21.535	0.777	BH	98
H3	Firth PML; NAVCO 1.3; Success (strict, binary); Moderator: Polity	Firth PML	NAVCO 1.3	Success (strict, binary)	Polity	2.478	0.703	BH	94
H3	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: CSO repression	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	CSO repression	-2.719	0.777	BH	64
H3	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: Civil liberties	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	Civil liberties	-56.839	0.703	BH	64
H3	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: Political liberty	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	Political liberty	-29.629	0.777	BH	64
H3	Cox PH; NAVCO 2.1; Hazard of success (CY, strict); Moderator: Polity	Cox PH	NAVCO 2.1	Hazard of success (CY, strict)	Polity	2.220	0.717	BH	62
H3	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: CSO repression	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	CSO repression	-3.863	0.777	BH	64
H3	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: Civil liberties	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	Civil liberties	-18.296	0.855	BH	64
H3	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: Political liberty	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	Political liberty	-20.644	0.788	BH	64
H3	Firth PML; NAVCO 2.1; Yearly success (CY, strict bin.); Moderator: Polity	Firth PML	NAVCO 2.1	Yearly success (CY, strict bin.)	Polity	0.586	0.855	BH	62

Appendix Table C10: Sensitivity to unobserved confounding

LPM approximation using semsemakr. RV is the partial R2 needed to reduce the estimate to zero.

Family	Moderator	Beta (LPM)	SE	p	RV	RV alpha	Bench R2y	Bench R2d	Adj. beta	N	Assessment
H2.1 Political	Domestic autonomy	2.216	1.587	0.166	0.135	---	0.056	0.001	2.120	101	Weak
H2.1 Political	State territorial control	0.157	0.115	0.175	0.133	---	0.058	0.001	0.150	101	Weak
H2.2 Economic	FDI	-0.107	0.493	0.828	0.024	---	0.071	0.002	-0.048	97	Fragile
H2.2 Economic	Trade	2.609	1.053	0.015	0.244	0.052	0.136	0.001	2.494	91	Moderate
H2.2 Economic	Aid	3.218	2.331	0.176	0.210	---	0.118	0.062	2.015	44	Moderate
H2.2 Economic	IO donor count	1.948	0.656	0.005	0.396	0.143	0.124	0.007	1.837	44	Robust
H3 Domestic	Repress	2.084	1.230	0.094	0.171	---	0.015	0.014	1.924	94	Moderate
H3 Domestic	CivLib	11.214	6.946	0.110	0.160	---	0.010	0.001	11.019	98	Moderate
H3 Domestic	Political Liberty	10.109	6.836	0.143	0.148	---	0.001	0.006	9.949	98	Weak
H3 Domestic	Polity	0.630	0.332	0.061	0.190	---	0.110	0.017	0.498	94	Moderate

Part VI. Power Diagnostics

Appendix Table C11: Power Diagnostics for Main-Paper Models

Reported power is the simulated rejection rate when the true focal effect is set equal to the fitted coefficient from the corresponding reported model. All values are drawn directly from the power diagnostic outputs.

H1 appears twice per reported model because the diagnostics report both the published squared-English coefficient test and the joint English-share test. Cox power rows that fail calibration checks are excluded from this reader-facing table. Weibull rows for the economic vulnerability and domestic-institution models are treated as diagnostics because the nominal Wald tests are not size-calibrated.

Power analysis method.

The reported power values are estimated by Monte Carlo simulation, not by closed-form analytic power formulas. Each power script uses the observed covariate distribution and the corresponding fitted model as the design template. For each model, the focal coefficient is set to a grid of effect-size multipliers (0, 0.25, 0.5, 1.0, and 2.0 times the fitted coefficient), simulated outcomes are generated, the original model is refit, and the focal p-value is recorded. Power is the share of simulated replications in which the focal test rejects at $\alpha = 0.05$ (Baranger et al. 2023).

The outcome simulator follows the estimator. Firth penalized logit rows simulate binary outcomes from Bernoulli draws using the fitted logit probability. Weibull accelerated-failure-time rows simulate event times from the fitted Weibull model and censor simulated event times at the observed campaign duration. Cox proportional-hazards power rows are excluded from this reader-facing table because the calibration audit found invalid or inflated false-positive rates under the current risk-set simulator.

The false-positive-rate column reports the Monte Carlo rejection rate when the true focal effect is set to zero. The fitted-effect-power column reports the rejection rate when the true focal effect equals the fitted coefficient. Minimum-detectable-effect diagnostics, where reported in the source files, are derived from the same simulation grid rather than from an analytic formula. The separate Weibull bootstrap calibration diagnoses small-sample false-positive behavior of the Weibull Wald tests; it is used to flag nominal-Wald rows, not to replace the Monte Carlo power estimates reported here.

Column notation.

Family identifies the hypothesis family; main table identifies the corresponding table in the manuscript; main-table model label gives the label used in that table; estimator records the fitted model class; and focal test identifies the coefficient or joint test for which power is reported.

Power at fitted effect is the simulated rejection rate at the nominal 0.05 level when the true focal effect equals the fitted coefficient from the reported model. False-positive rate is the simulated rejection rate at the nominal 0.05 level when the true focal effect is set to zero.

Status / note flags rows that should be treated as diagnostics rather than size-calibrated power estimates, including nominal-Wald Weibull diagnostics.

Family	Main table	Main-table model label	Estimator	Focal test	Power at fitted effect	False-positive rate	Status / note
H1	Table 1	M1	Weibull AFT	English share squared	0.409	0.032	
H1	Table 1	M1	Weibull AFT	English share + squared English share	0.282	0.042	
H1	Table 1	M2	Firth PML	English share squared	0.119	0.026	
H1	Table 1	M2	Firth PML	English share + squared English share	0.063	0.022	
H1	Table 1	M5	Firth PML	English share squared	0.114	0.001	
H1	Table 1	M5	Firth PML	English share + squared English share	0.048	0.000	
H2.1	Table 2	M1	Weibull AFT	English share x Domestic Autonomy	0.189	0.071	
H2.1	Table 2	M2	Weibull AFT	English share x State Authority Over Territory	0.113	0.070	
H2.1	Table 2	M3	Firth PML	English share x Domestic Autonomy	0.235	0.044	
H2.1	Table 2	M4	Firth PML	English share x State Authority Over Territory	0.197	0.060	
H2.1	Table 2	M9	Firth PML	English share x Domestic Autonomy	0.167	0.048	
H2.1	Table 2	M10	Firth PML	English share x State Authority Over Territory	0.059	0.043	
H2.2	Table 3	M1	Weibull AFT	English share x FDI inflows	0.393	0.078	Weibull nominal-Wald diagnostic; not size-calibrated.
H2.2	Table 3	M2	Weibull AFT	English share x Trade openness	0.666	0.082	Weibull nominal-Wald diagnostic; not size-calibrated.
H2.2	Table 3	M3	Weibull AFT	English share x Foreign aid	0.121	0.110	Weibull nominal-Wald diagnostic; not size-calibrated.
H2.2	Table 3	M4	Weibull AFT	English share x IO donor count	0.203	0.131	Weibull nominal-Wald diagnostic; not size-calibrated.
H2.2	Table 3	M5	Firth PML	English share x FDI inflows	0.049	0.040	
H2.2	Table 3	M6	Firth PML	English share x Trade openness	0.673	0.042	
H2.2	Table 3	M7	Firth PML	English share x Foreign aid	0.217	0.043	
H2.2	Table 3	M8	Firth PML	English share x IO donor count	0.695	0.043	
H3	Table 4	M1	Weibull AFT	English share x CSO repression	0.436	0.086	Weibull nominal-Wald diagnostic; not size-calibrated.
H3	Table 4	M2	Weibull AFT	English share x Civil liberties	0.314	0.081	Weibull nominal-Wald diagnostic; not size-calibrated.
H3	Table 4	M3	Weibull AFT	English share x Political liberty	0.377	0.079	Weibull nominal-Wald diagnostic; not size-calibrated.
H3	Table 4	M4	Weibull AFT	English share x Polity	0.080	0.089	Weibull nominal-Wald diagnostic; not size-calibrated.
H3	Table 4	M5	Firth PML	English share x CSO repression	0.475	0.042	
H3	Table 4	M6	Firth PML	English share x Civil liberties	0.461	0.036	
H3	Table 4	M7	Firth PML	English share x Political liberty	0.470	0.040	

H3	Table 4	M8	Firth PML	English share x Polity	0.677	0.048	
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Implications for power analysis.

Because the analysis is constrained to a substantively bounded set of campaigns, we do not treat the conventional 0.80 benchmark as a sharp pass/fail threshold. Instead, the table reports per-test Monte Carlo power as a graded design-sensitivity diagnostic: the question is how much information the observed design contains about the fitted focal effect, not whether the design crosses a universal adequacy line (Bacchetti 2010; Lakens 2022).

On that graded reading, H1 remains weak after Cox diagnostics that fail calibration checks are excluded: the valid Weibull and Firth rows have low to modest fitted-effect power. The excluded Cox rows are retained only in the inference-diagnostics appendix because the audit flags the simulated risk-set process as invalid for reader-facing power analysis.

H2.1 is the weakest family among the main interaction tests. After excluding Cox rows that fail calibration checks, the Table 2 power evidence comes from the Weibull and Firth specifications, including the NAVCO 2.1 yearly Firth rows. Weak or unstable H2.1 findings should therefore be interpreted as both substantively weak and statistically underpowered; the diagnostics cannot rule out meaningful false negatives for that hypothesis. This is consistent with broader evidence that interaction-heavy quantitative political science designs often have limited power in small samples (Arel-Bundock et al. 2026; Baranger et al. 2023).

For H2.2 and H3, the picture is more informative but still not uniformly high-powered. Some valid Firth PML interaction models show moderate raw fitted-effect power, especially H2.2 trade openness (0.673), H2.2 IO donor count (0.695), and H3 polity (0.677), while the H3 repression, civil-liberties, and political-liberty Firth models fall around 0.46 to 0.48. Other focal tests remain weak, including H2.2 FDI inflows (0.049) and H2.2 foreign aid (0.217). The Weibull rows for H2.2 and H3 should be read only as nominal-Wald diagnostics rather than size-calibrated power estimates.

The resulting interpretation is mixed but not empty. The design retains nontrivial ability to detect the larger fitted interaction patterns in several H2.2 and H3 Firth specifications, while H1 and especially H2.1 remain fragile. The appendix table should therefore be read as a transparency check that distinguishes very weak hypothesis families from more informative, but still small-N-constrained, interaction tests.

References for power note.

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Part VII. Inference Diagnostics

Appendix Table C12: Inference Diagnostics for Survival and Power Analyses

This table consolidates inference. It separates Cox proportional-hazards assumption checks, Cox power rows excluded from reader-facing power summaries, and Weibull bootstrap calibration diagnostics. These rows are diagnostics and provenance checks; they do not replace the main coefficient tables.

Panel A. Cox proportional-hazards diagnostics

Proportional-hazards diagnostics are reported for first-event Cox rows where available. Recurrent-event robustness rows are explicitly marked when the diagnostic is not available from the inference-diagnostics output.

Family	Main table	Model	Role	Moderator	Response / estimand	Cox estimand	N	Events	Global PH p	Focal PH p	Diagnostic status	Note
H1	Table 1	Main panel; Hazard of first concession (CY)	Main panel		Hazard of first concession (CY)	first_concession	59	13	0.496	0.312	No cox.zph flag	
H1	Table 1	Appendix strict-success; Hazard of success (CY, strict)	Appendix strict-success		Hazard of success (CY, strict)	strict_success	65	8	0.691	0.629	No cox.zph flag	
H1	Table 1	Appendix robustness; Hazard of concession (Cox PH, campaign-lvl)	Appendix robustness		Hazard of concession (Cox PH, campaign-lvl)	campaign_level_concession	100	45	0.537	0.249	No cox.zph flag	
H1	Table 1	Recurrent-event robustness; Hazard of concession (CY, recurrent-event)	Recurrent-event robustness		Hazard of concession (CY, recurrent-event)	recurrent_event_concession	65	15	NA	NA	Not reported for recurrent-event robustness	Recurrent-event Cox robustness estimand; first-event PH diagnostic not reported
H2.1	Table 2	Main panel; Hazard of first concession (CY); Moderator: Domestic autonomy	Main panel	Domestic autonomy	Hazard of first concession (CY)	first_concession	59	13	0.793	0.463	No cox.zph flag	
H2.1	Table 2	Appendix strict-success; Hazard of success (CY, strict); Moderator: Domestic autonomy	Appendix strict-success	Domestic autonomy	Hazard of success (CY, strict)	strict_success	65	8	0.884	0.687	No cox.zph flag	

H2.1	Table 2	Main panel; Hazard of first concession (CY); Moderator: State territorial control	Main panel	State territorial control	Hazard of first concession (CY)	first_concession	59	13	0.888	0.314	No cox.zph flag	
H2.1	Table 2	Appendix strict-success; Hazard of success (CY, strict); Moderator: State territorial control	Appendix strict-success	State territorial control	Hazard of success (CY, strict)	strict_success	65	8	0.861	0.845	No cox.zph flag	
H2.1	Table 2	Recurrent-event robustness; Hazard of concession (CY, recurrent-event); Moderator: Domestic autonomy	Recurrent-event robustness	Domestic autonomy	Hazard of concession (CY, recurrent-event)	recurrent_event_concession	65	15	NA	NA	Not reported for recurrent-event robustness	Recurrent-event robustness estimand; first-event PH diagnostic not reported
H2.1	Table 2	Recurrent-event robustness; Hazard of concession (CY, recurrent-event); Moderator: State territorial control	Recurrent-event robustness	State territorial control	Hazard of concession (CY, recurrent-event)	recurrent_event_concession	65	15	NA	NA	Not reported for recurrent-event robustness	Recurrent-event Cox robustness estimand; first-event PH diagnostic not reported
H2.2	Table 3	Appendix robustness; Hazard of first success (CY); Moderator: FDI inflows	Appendix robustness	FDI inflows	Hazard of first success (CY)	first_success_broad	55	13	0.623	0.315	No cox.zph flag	
H2.2	Table 3	Appendix strict-success; Hazard of success (CY, strict); Moderator: FDI inflows	Appendix strict-success	FDI inflows	Hazard of success (CY, strict)	strict_success	61	8	0.798	0.192	No cox.zph flag	
H2.2	Table 3	Appendix robustness; Hazard of first success (CY); Moderator: Trade openness	Appendix robustness	Trade openness	Hazard of first success (CY)	first_success_broad	54	12	0.166	0.252	No cox.zph flag	
H2.2	Table 3	Appendix strict-success; Hazard of success (CY, strict); Moderator: Trade openness	Appendix strict-success	Trade openness	Hazard of success (CY, strict)	strict_success	60	7	0.796	0.709	No cox.zph flag	
H2.2	Table 3	Appendix robustness; Hazard of first success (CY); Moderator: Foreign aid	Appendix robustness	Foreign aid	Hazard of first success (CY)	first_success_broad	53	11	0.877	0.391	No cox.zph flag	
H2.2	Table 3	Appendix strict-success; Hazard of success (CY, strict); Moderator: Foreign aid	Appendix strict-success	Foreign aid	Hazard of success (CY, strict)	strict_success	59	8	0.852	0.724	No cox.zph flag	
H2.2	Table 3	Appendix robustness; Hazard of first success (CY); Moderator: IO donor count	Appendix robustness	IO donor count	Hazard of first success (CY)	first_success_broad	53	11	0.794	0.779	No cox.zph flag	
H2.2	Table 3	Appendix strict-success; Hazard of success (CY, strict); Moderator: IO donor count	Appendix strict-success	IO donor count	Hazard of success (CY, strict)	strict_success	59	8	0.956	0.524	No cox.zph flag	
H2.2	Table 3	Recurrent-event robustness; Hazard of recurrent campaign success (CY); Moderator: FDI inflows	Recurrent-event robustness	FDI inflows	Hazard of recurrent campaign success (CY)	recurrent_event_success_broad	61	15	NA	NA	Not reported for recurrent-event robustness	Recurrent-event broad-success Cox robustness estimand; first-event PH diagnostic not reported
H2.2	Table 3	Recurrent-event robustness; Hazard of recurrent campaign success	Recurrent-event robustness	Trade openness	Hazard of recurrent campaign success (CY)	recurrent_event_success_broad	60	14	NA	NA	Not reported for recurrent-event robustness	Recurrent-event broad-success Cox robustness estimand; first-

		(CY); Moderator: Trade openness										event PH diagnostic not reported
H2.2	Table 3	Recurrent-event robustness; Hazard of recurrent campaign success (CY); Moderator: Foreign aid	Recurrent-event robustness	Foreign aid	Hazard of recurrent campaign success (CY)	recurrent_event_success_broad	59	13	NA	NA	Not reported for recurrent-event robustness	Recurrent-event broad-success Cox robustness estimand; first-event PH diagnostic not reported
H2.2	Table 3	Recurrent-event robustness; Hazard of recurrent campaign success (CY); Moderator: IO donor count	Recurrent-event robustness	IO donor count	Hazard of recurrent campaign success (CY)	recurrent_event_success_broad	59	13	NA	NA	Not reported for recurrent-event robustness	Recurrent-event broad-success Cox robustness estimand; first-event PH diagnostic not reported
H3	Table 4	Appendix robustness; Hazard of first concession (CY); Moderator: CSO repression	Appendix robustness	CSO repression	Hazard of first concession (CY)	first_concession	58	13	0.690	0.816	No cox.zph flag	
H3	Table 4	Appendix strict-success; Hazard of success (CY, strict); Moderator: CSO repression	Appendix strict-success	CSO repression	Hazard of success (CY, strict)	strict_success	64	8	0.898	0.765	No cox.zph flag	
H3	Table 4	Appendix robustness; Hazard of first concession (CY); Moderator: Civil liberties	Appendix robustness	Civil liberties	Hazard of first concession (CY)	first_concession	58	13	0.413	0.810	No cox.zph flag	
H3	Table 4	Appendix strict-success; Hazard of success (CY, strict); Moderator: Civil liberties	Appendix strict-success	Civil liberties	Hazard of success (CY, strict)	strict_success	64	8	0.864	0.365	No cox.zph flag	
H3	Table 4	Appendix robustness; Hazard of first concession (CY); Moderator: Political liberty	Appendix robustness	Political liberty	Hazard of first concession (CY)	first_concession	58	13	0.768	0.922	No cox.zph flag	
H3	Table 4	Appendix strict-success; Hazard of success (CY, strict); Moderator: Political liberty	Appendix strict-success	Political liberty	Hazard of success (CY, strict)	strict_success	64	8	0.845	0.888	No cox.zph flag	
H3	Table 4	Appendix robustness; Hazard of first concession (CY); Moderator: Polity	Appendix robustness	Polity	Hazard of first concession (CY)	first_concession	56	13	0.202	0.520	No cox.zph flag	
H3	Table 4	Appendix strict-success; Hazard of success (CY, strict); Moderator: Polity	Appendix strict-success	Polity	Hazard of success (CY, strict)	strict_success	62	8	0.934	0.963	No cox.zph flag	
H3	Table 4	Recurrent-event robustness; Hazard of concession (CY, recurrent-event); Moderator: CSO repression	Recurrent-event robustness	CSO repression	Hazard of concession (CY, recurrent-event)	recurrent_event_concession	64	15	NA	NA	Not reported for recurrent-event robustness	Recurrent-event Cox robustness estimand; first-event PH diagnostic not reported
H3	Table 4	Recurrent-event robustness; Hazard of concession (CY, recurrent-event); Moderator: Civil liberties	Recurrent-event robustness	Civil liberties	Hazard of concession (CY, recurrent-event)	recurrent_event_concession	64	15	NA	NA	Not reported for recurrent-event robustness	Recurrent-event Cox robustness estimand; first-event PH diagnostic not reported
H3	Table 4	Recurrent-event robustness; Hazard of concession (CY, recurrent-event); Moderator: Political liberty	Recurrent-event robustness	Political liberty	Hazard of concession (CY, recurrent-event)	recurrent_event_concession	64	15	NA	NA	Not reported for recurrent-event robustness	Recurrent-event Cox robustness estimand; first-event PH diagnostic not reported
H3	Table 4	Recurrent-event robustness; Hazard of concession (CY, recurrent-event)	Recurrent-event robustness	Polity	Hazard of concession (CY, recurrent-event)	recurrent_event_concession	62	15	NA	NA	Not reported for recurrent-event robustness	Recurrent-event Cox robustness estimand; first-

		recurrent-event); Moderator: Polity										event PH diagnostic not reported
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